Question ID: 105ea6de

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-Text Connections	Hard

Question

Text 1

Growth in the use of novel nanohybrids—materials created from the conjugation of multiple distinct nanomaterials, such as iron oxide and gold nanomaterials conjugated for use in magnetic imaging—has outpaced studies of nanohybrids' environmental risks. Unfortunately, risk evaluations based on nanohybrids' constituents are not reliable: conjugation may alter constituents' physiochemical properties such that innocuous nanomaterials form a nanohybrid that is anything but.

Text 2

The potential for enhanced toxicity of nanohybrids relative to the toxicity of constituent nanomaterials has drawn deserved attention, but the effects of nanomaterial conjugation vary by case. For instance, it was recently shown that a nanohybrid of silicon dioxide and zinc oxide preserved the desired optical transparency of zinc oxide nanoparticles while mitigating the nanoparticles' potential to damage DNA.

Based on the texts, how would the author of Text 2 most likely respond to the assertion in the underlined portion of Text 1?

Answer

A. By concurring that the risk described in Text 1 should be evaluated but emphasizing that the risk is more than offset by the potential benefits of nanomaterial conjugation

B. By arguing that the situation described in Text 1 may not be representative but conceding that the effects of nanomaterial conjugation are harder to predict than researchers had expected

C. By denying that the circumstance described in Text 1 is likely to occur but acknowledging that many aspects of nanomaterial conjugation are still poorly understood

D. By agreeing that the possibility described in Text 1 is a cause for concern but pointing out that nanomaterial conjugation does not inevitably produce that result

Correct Answer: D

Rationale

Choice D is the best answer. The author of Text 2 acknowledges that nanohybrids may be more toxic than their constituent parts, but also provides an example of a nanohybrid that has reduced toxicity compared to its components: silicon dioxide and zinc oxide together have all the benefits of zinc oxide nanoparticles without any of the DNA harm zinc oxide has on its own.

Choice A is incorrect. While the author of Text 2 gives an example of a nanohybrid that isn't as toxic as its constituent parts, they don't argue that the benefit outweighs the risk. They merely argue that "the effects of nanomaterial conjugation vary by case." Choice B is incorrect. The author of Text 2 states that the effects of nanomaterial conjugation "vary by case," and that the attention that their potential toxicity has drawn is warranted. If the situation in Text 1 weren't representative, then there would be less attention to the potential danger of these materials. Furthermore, neither passage suggests that researchers had expected that they could predict the effects of nanomaterial conjugation. Choice C is incorrect. The author of Text 2 agrees that the potential toxicity of nanohybrids "has drawn deserved attention," so they aren't denying the problem.